Failure of inverse closure for Star of David matrices

Principia Math in Progress

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Task 1. Resolve Sun's inverse-closure conjecture. Conjecture 2.6 of [1] asks whether every invertible $A \in \mathrm{SDR}_m$ has $A^{-1} \in \mathrm{SDR}_m$. We give a counterexample simultaneously for every $m \geq 3$, even with $A \in \mathrm{SDR}_\infty$. We use Sun's source definition: SDR means *Star of David rule*. The added phrase "sign-determined Riordan matrices" in MathDB is inconsistent with that definition. No Riordan-array restriction is imposed in the source conjecture; our result resolves that original assertion, and makes no assertion about a separately restricted class.

For an infinite lower triangular matrix X, with zero entries above the diagonal, put

$$L_{p,r}(n,k;X) = \prod_{i=0}^{r} X_{n+i,k+r-i} \prod_{i=0}^{p-r-1} X_{n+p-i,k+r+i+1},$$

$$R_{p,r}(n,k;X) = \prod_{i=0}^{r} X_{n+p-i,k+p-r+i} \prod_{i=0}^{p-r-1} X_{n+i,k+p-r-i-1}.$$

Definition 1.1 in [1] requires $L_{p,r} = R_{p,r}$ for all $n \geq k \geq 0$, $2 \leq p \leq m-1$, and $0 \leq r \leq p-1$. Membership in SDR_∞ means these identities hold for every $p \geq 2$.

Lemma 1. *Let M be a lower triangular $d \times d$ matrix satisfying all the displayed identities whose row indices lie in $\{0, \ldots, d-1\}$. Then $M \oplus I \in \mathrm{SDR}_\infty$, where I is the infinite identity.*

Proof. Each product uses every row $n, \ldots, n+p$ and every column $k, \ldots, k+p$ exactly once. If $n > k$ and $n + p \geq d$, row $n+p$ belongs to the identity tail. Its only nonzero entry is in column $n+p$, exceeding all the columns in either product. Consequently both products vanish. If $n + p < d$, the assumed finite identities apply.

It remains to handle $n = k$. For $r \geq 1$, the first factor $X_{n,k+r}$ of L is above the diagonal; for $r = 0$, the factor with $i = p-1$ in its second product is $X_{n+1,k+p} = 0$. In R, if $r \leq p-2$, its second product contains $X_{n,k+p-r-1} = 0$; if $r = p-1$, its first product contains $X_{n+1,k+p} = 0$. Thus both sides vanish here too. □

Consider

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 2 & 1 \end{pmatrix}, \qquad A = M \oplus I.$$

All identities within M hold. To check this without any unbounded computation, cases $n = k$ were proved in the lemma. For $n > k$ and $n + p \leq 4$, the only possibilities are the following:

p	(n,k)	r	L	R
2	$(1,0)$	0	$1 \cdot 0 \cdot 1 = 0$	$1 \cdot 1 \cdot 0 = 0$
2	$(2,0)$	0	$0 \cdot 1 \cdot 1 = 0$	$0 \cdot 1 \cdot 0 = 0$
2	$(2,1)$	0	$1 \cdot 0 \cdot 1 = 0$	$2 \cdot 1 \cdot 0 = 0$
3	$(1,0)$	0	$0 \quad (M_{2,3} = 0)$	$0 \quad (M_{1,2} = 0)$
3	$(1,0)$	1	$0 \quad (M_{2,0} = 0)$	$0 \quad (M_{2,0} = 0)$

For $p = 2, r = 1$, the two sides are the $r = 0$ sides interchanged. For $p = 3, r = 2$, they are the $r = 0$ sides interchanged. There are no further cases with $n > k$. The lemma proves $A \in \mathrm{SDR}_\infty$.

Theorem 1. *The matrix A belongs to every SDR_m, but A^{-1} belongs to none of them, for $m \geq 3$.*

Proof. The matrix is unit lower triangular and

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 \\ -1 & 1 & -1 & 1 & 0 \\ 3 & -3 & 2 & -2 & 1 \end{pmatrix}.$$

Direct multiplication on both sides gives the identity. Thus $B = A^{-1} = M^{-1} \oplus I$ exists as an infinite row-finite matrix. But with $(p, n, k, r) = (2, 2, 0, 0)$,

$$L = B_{2,0}B_{4,1}B_{3,2} = 1(-3)(-1) = 3, \qquad R = B_{4,2}B_{2,1}B_{3,0} = 2(-1)(-1) = 2.$$

This violates the order-three rule, which is included in every SDR_m with $m \geq 3$. □

Verification and scope. The supplied script checks all 25 finite identities, the displayed matrix operations, and the exact failed identity. It also checks a larger truncation as a diagnostic. Lemma 1 supplies the proof for the entire infinite matrix; the diagnostic is not used to infer an infinite assertion. Zero entries are allowed in Definition 1.1 and in the selected conjecture. The source explicitly imposes a nonzero-entry hypothesis in other theorems, including Theorem 2.10, but not here. The same matrix also supplies a counterexample to the distinct contiguous-minor closure question, treated separately as MathDB 317354. The operations and failed identities are different.

Task 2. Minimize the counterexample. Determine the smallest leading block size permitting failure of inverse closure, and classify examples of that size up to diagonal rescaling.

Task 3. Classify support patterns. Find conditions on the zero pattern of an invertible Star of David matrix that guarantee its inverse satisfies the same rules.

Task 4. Study nonvanishing arrays. Determine whether every nowhere-zero Star of David matrix admits a row–column–diagonal factorization, and derive consequences for inversion.

Task 5. Analyze the Riordan intersection. Investigate inverse closure within the intersection of Sun's SDR_m and the ordinary Riordan group.

References

[1] Y. Sun, *The Star of David rule*, Linear Algebra Appl. **429** (2008), 1954–1961, doi:10.1016/j.laa.2008.05.027; arXiv:0805.1277v1. Source references in this note use the actual seven-page arXiv PDF.